

From folder to finished report: a three-day walkthrough

Consumer Decision-Making Analysis in R

The Science Repository — workshop edition

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1 Why this document exists

This is the **one orchestration file** for the workshop. It walks you through the analysis end-to-end and shows how the folder structure you built on Day 1 removes duplication on Days 2 and 3.

The rule for this document: **nothing is redefined here**. Variable names, cleaning logic, models, plots — they all live in the **R/** folder. This walkthrough just **source()**s them and tells the story.

That is the whole point. Change a model in `R/functions/analysis.R` and the website, the PDF, the manuscript, and the Shiny app all update the next time you render. No copy-paste.

1.1 How to render

One command, three folders:

```
quarto render
```

Output	Lands in
00_walkthrough.html (and the rest of the site)	<code>docs/</code> → GitHub Pages
walkthrough.pdf	<code>manuscript/output/</code>
walkthrough.docx	<code>manuscript/output/</code>

The HTML/PDF/DOCX split is one source file, three formats. A post-render script (`scripts/post_render.R`) moves the PDF and DOCX out of `docs/` and into `manuscript/output/` so the site folder stays HTML-only.

PDF requires a LaTeX engine. One-time setup:

```
install.packages("tinytex")
tinytex::install_tinytex()
```

To render only a subset of formats (e.g., if tinytex isn't installed):

```
quarto render reports/00_walkthrough.qmd --to html
quarto render reports/00_walkthrough.qmd --to html,docx
```

2 Day 1 — The scaffold

What you built yesterday is a **project**, not a folder. The difference:

```
folder_overview <- tibble::tribble(
  ~Folder,      ~Purpose,
  "R/",          "Code: numbered scripts + reusable functions.",
  "data/mock/",  "Synthetic, committed, LLM-safe data.",
  "data/raw/",   "Real data. Gitignored.",
  "data/processed/", "Output of cleaning. Reproducible, gitignored.",
  "reports/",    "Quarto sources (this file).",
  "docs/",       "Rendered website. GitHub Pages target.",
  "manuscript/", "LaTeX paper.",
  "shiny/",      "Interactive app.",
  "references/", "One .bib file used by both Quarto and LaTeX.",
  "figures/",    "Plots saved by scripts, shared by every output."
)
knitr::kable(folder_overview)
```

Table 2: What every folder in this project does.

Folder	Purpose
R/	Code: numbered scripts + reusable functions.
data/mock/	Synthetic, committed, LLM-safe data.
data/raw/	Real data. Gitignored.
data/processed/	Output of cleaning. Reproducible, gitignored.
reports/	Quarto sources (this file).
docs/	Rendered website. GitHub Pages target.
manuscript/	LaTeX paper.
shiny/	Interactive app.
references/	One .bib file used by both Quarto and LaTeX.
figures/	Plots saved by scripts, shared by every output.

You also wrote a smart `.gitignore` so real data can never be committed by accident, and an `.Renviron.example` with one switch:

```
DATA_MODE=mock    # default - read from data/mock/
DATA_MODE=real    # only when you mean it - read from data/raw/
```

Everything that follows in this document reads through `data_path()`. Flip `DATA_MODE` and the entire analysis re-runs on a different dataset, with no edits to any script.

3 Day 2 — From raw to results

3.1 Load the raw data

```
raw <- load_raw_consumer_data("consumer_data_raw.csv")
dplyr::glimpse(raw)
```

```
Rows: 300
Columns: 26
$ ParticipantID      <chr> "P001", "P002", "P003", "P004", "P005", "P006", ~
$ Condition          <chr> "scarcity_frame", "combined_frame", "social_pro~
$ Age                <dbl> 36, 63, 41, 51, 38, 46, 46, 43, 40, 27, 58, 58, ~
$ Gender             <chr> "female", "M", "Female", "non-binary", "W", "ma~
$ IncomeGroup        <chr> "low", "low", "medium", "medium", "medium", "me~
$ ProductCategory    <chr> "food", "clothing", "services", "electronics", ~
$ ps_1               <dbl> 5, 2, 5, 5, 2, 4, 6, 4, 4, 3, 3, 4, 6, 2, 2, 5, ~
$ ps_2               <dbl> 7, 4, 5, 4, 3, 5, 6, 5, 5, 1, 3, 2, 6, 4, 2, 2, ~
$ ps_3               <dbl> 7, 3, 3, 4, 3, 4, 5, 3, 4, 2, 5, 3, 4, 5, 3, 5, ~
$ sc_1               <dbl> 3, 4, 6, 3, 4, 4, 4, 4, 4, 5, 5, 4, 6, 4, 4, 4, ~
$ sc_2               <dbl> 3, 4, 6, 2, 4, 4, 3, 4, 6, 4, 5, 4, 7, 4, 5, 3, ~
$ sc_3               <dbl> 3, 4, 5, 5, 4, 4, 4, 5, 4, 5, 5, 4, 6, 5, 6, 4, ~
$ bt_1               <dbl> 6, 4, 2, 4, NA, 3, 6, 4, 4, 3, 3, 6, 7, 2, 4, 6~
$ bt_2               <dbl> 6, 4, 1, 3, 5, 3, 6, 4, 5, 1, 4, 4, 5, 4, 3, 5, ~
$ bt_3_r             <dbl> 2, 2, 5, 6, 1, 4, 3, 3, 5, 5, 5, 3, 1, 5, 5, 4, ~
$ pv_1               <dbl> 4, 5, 3, 3, 4, 3, 4, 3, 4, 3, 4, 3, 5, 3, 4, 5, ~
$ pv_2               <dbl> 5, 3, 5, 4, 4, 2, 3, 4, 4, 5, 4, 5, 6, 3, 5, 4, ~
```

```

$ pv_3 <dbl> 5, 4, 2, 3, 3, 3, 3, 4, 5, 3, 5, 3, 6, 4, 5, 4,~
$ perceived_quality <dbl> 6, 3, 3, 4, 4, 3, 5, 4, 4, 3, 4, 4, 5, 2, 4, 4,~
$ perceived_social_norm <dbl> 4, 5, 5, 4, 5, 5, 6, 4, 3, 4, 5, 4, 6, 2, 5, 5,~
$ pi_1 <dbl> 3, 4, 4, 2, NA, 5, 4, 3, 3, 3, 2, 5, 4, 5, 5, 5~
$ pi_2 <dbl> 2, 4, 4, 2, 3, 4, 4, 4, 2, 4, 4, 3, 4, 3, 4, 5,~
$ pi_3 <dbl> 5, 3, 4, 4, 3, 5, 4, 3, 3, 3, 3, 5, 5, 5, 6, 5,~
$ willingness_to_pay <dbl> 35.31, 39.74, 39.68, 44.55, 23.99, 48.11, 41.49~
$ choice <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1,~
$ number_products_bought <dbl> 2, 4, 3, 1, 3, 2, 0, 2, 0, 4, 1, 2, 4, 1, 2, 3,~

```

The CSV is full of realistic mess: mixed-case column names, trailing whitespace, -99 missing codes, inconsistent gender labels, a reverse-scored item, scattered NAs. See [data/mock/codebook.md](#) for the full inventory.

3.2 Clean it

The cleaning logic lives in [R/functions/data_loading.R](#) so the cleaning script, this walkthrough, and the Shiny app all share one implementation:

```

clean <- clean_consumer_data(raw)
dplyr::glimpse(dplyr::select(clean, participant_id, condition, gender,
                             income_group, purchase_intention,
                             perceived_value, price_sensitivity))

```

```

Rows: 300
Columns: 7
$ participant_id <chr> "P001", "P002", "P003", "P004", "P005", "P006", "P0~
$ condition <fct> scarcity_frame, combined_frame, social_proof_frame,~
$ gender <fct> woman, man, woman, non_binary, woman, man, man, wom~
$ income_group <ord> low, low, medium, medium, medium, medium, medium, h~
$ purchase_intention <dbl> 3.333333, 3.666667, 4.000000, 2.666667, 3.000000, 4~
$ perceived_value <dbl> 4.666667, 4.000000, 3.333333, 3.333333, 3.666667, 2~
$ price_sensitivity <dbl> 6.333333, 3.000000, 4.333333, 4.333333, 2.666667, 4~

```

What `clean_consumer_data()` does (see source for the details): standardises column names with `janitor::clean_names()`, trims whitespace, recodes gender, turns -99 into NA, reverses `bt_3_r`, and averages each item set into a scale score.

3.3 Descriptives

```

desc <- descriptives_by_condition(clean)
knitr::kable(desc, digits = 2)

```

Table 3: Outcomes by experimental condition.

condition	n	mean_pi	sd_pi	mean_pv	sd_pv	mean_wtp	prop_choice_yes
control	50	4.03	1.13	4.09	1.07	40.12	0.40
scarcity_frame	75	4.20	1.01	4.12	1.07	43.05	0.35

Table 3: Outcomes by experimental condition.

condition	n	mean_pi	sd_pi	mean_pv	sd_pv	mean_wtp	prop_choice_yes
sustainability_frame	67	4.35	1.06	4.51	1.04	43.83	0.52
social_proof_frame	53	4.28	1.19	4.25	0.93	44.35	0.40
combined_frame	55	4.60	0.96	4.88	1.04	48.55	0.47

3.4 Visual first, model second

```
plot_pi_by_condition(clean)
```

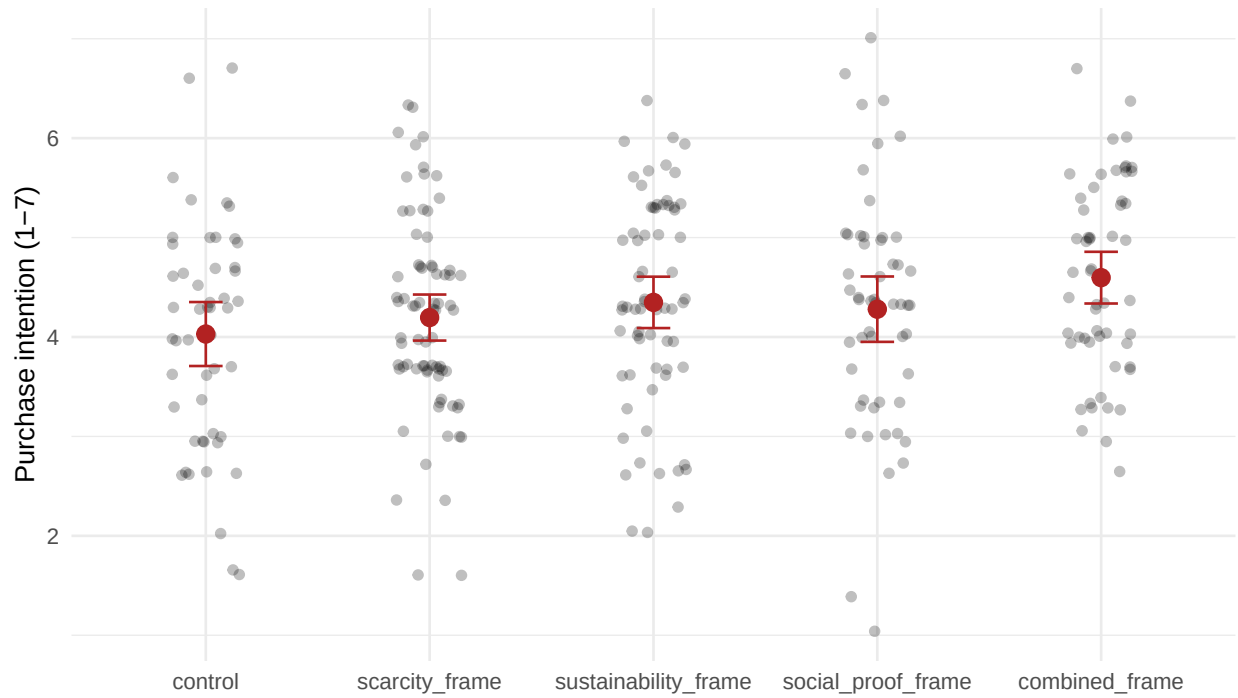


Figure 1: Purchase intention by condition. Red points are condition means; bars are 95% CIs.

`combined_frame` shows a clear bump over `control`. The scatter below suggests why: the conditions also raise `perceived_value`, and perceived value strongly predicts purchase intention.

```
plot_mediation_scatter(clean)
```

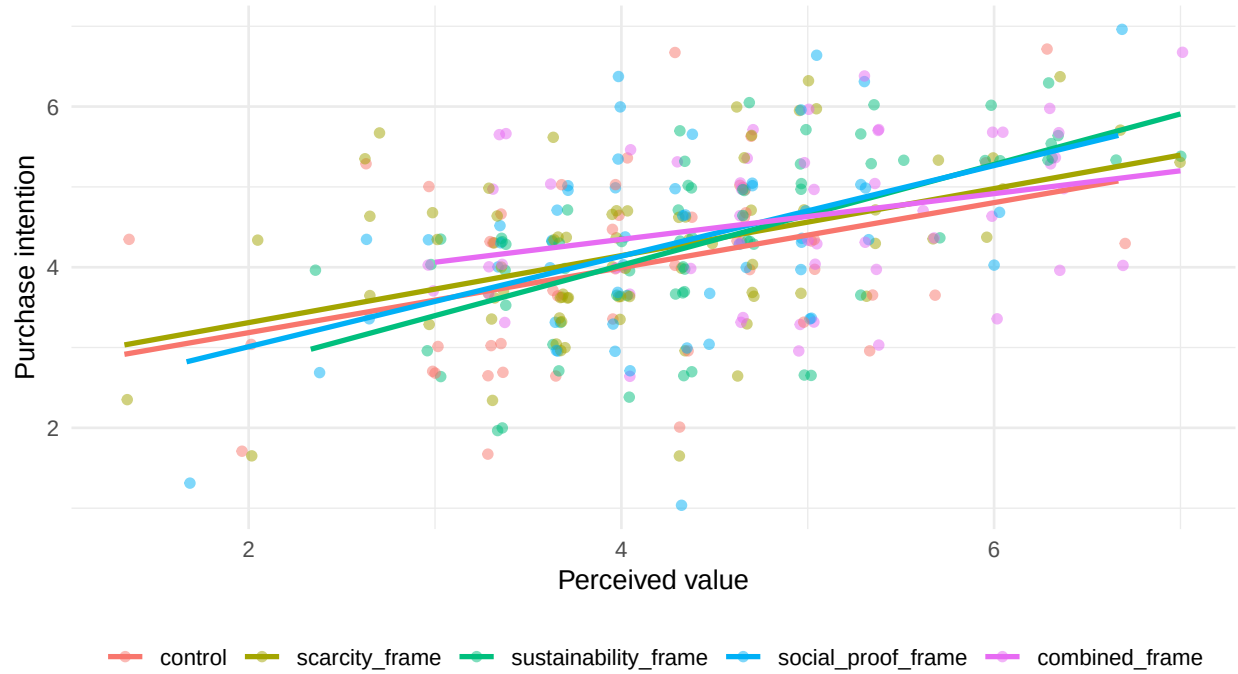


Figure 2: Perceived value (mediator) versus purchase intention (outcome), by condition.

3.5 Main effect: condition predicts purchase intention

```
fit_main <- fit_main_model(clean)
knitr::kable(broom::tidy(fit_main, conf.int = TRUE), digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	4.085	0.254	16.111	0.000	3.586	4.584
conditionscarcity_frame	0.140	0.204	0.685	0.494	-0.262	0.541
conditionsustainability_frame	0.311	0.206	1.508	0.133	-0.095	0.716
conditionsocial_proof_frame	0.240	0.218	1.102	0.271	-0.189	0.669
conditioncombined_frame	0.582	0.218	2.675	0.008	0.154	1.010
age	-0.001	0.006	-0.180	0.858	-0.012	0.010
genderman	-0.032	0.137	-0.233	0.816	-0.301	0.237
gendernon_binary	-0.465	0.342	-1.361	0.175	-1.137	0.208
genderprefer_not_to_say	0.124	0.297	0.418	0.676	-0.460	0.709
income_group.L	-0.175	0.130	-1.353	0.177	-0.431	0.080
income_group.Q	0.021	0.104	0.201	0.841	-0.184	0.226

3.6 Mediation: condition → perceived value → purchase intention

Three regressions tell the conceptual mediation story (Baron and Kenny 1986; Hayes 2017):

```
med <- fit_mediation_models(clean)
```

```
dplyr::bind_rows(
  total      = broom::tidy(med$total, conf.int = TRUE),
  mediator    = broom::tidy(med$a,      conf.int = TRUE),
  outcome_w_m = broom::tidy(med$b,      conf.int = TRUE),
  .id = "model"
) |>
dplyr::filter(grepl("condition|perceived_value", term)) |>
knitr::kable(digits = 3)
```

model	term	estimate	std.error	statistic	p.value	conf.low	conf.high
total	conditionscarcity_frame	0.140	0.204	0.685	0.494	-0.262	0.541
total	conditionsustainability_frame	0.311	0.206	1.508	0.133	-0.095	0.716
total	conditionsocial_proof_frame	0.240	0.218	1.102	0.271	-0.189	0.669
total	conditioncombined_frame	0.582	0.218	2.675	0.008	0.154	1.010
mediator	conditionscarcity_frame	0.045	0.196	0.227	0.820	-0.342	0.431
mediator	conditionsustainability_frame	0.483	0.199	2.433	0.016	0.092	0.874
mediator	conditionsocial_proof_frame	0.196	0.210	0.934	0.351	-0.217	0.609
mediator	conditioncombined_frame	0.849	0.210	4.050	0.000	0.436	1.262
outcome_w_m	conditionscarcity_frame	0.119	0.183	0.650	0.516	-0.242	0.480
outcome_w_m	conditionsustainability_frame	0.090	0.187	0.483	0.630	-0.278	0.459
outcome_w_m	conditionsocial_proof_frame	0.151	0.196	0.767	0.443	-0.236	0.537
outcome_w_m	conditioncombined_frame	0.195	0.201	0.968	0.334	-0.202	0.591
outcome_w_m	perceived_value	0.456	0.056	8.154	0.000	0.346	0.566

The pattern to read for in the table: the condition coefficients in `outcome_w_m` are **smaller** than in `total`, and `perceived_value` is a significant predictor — that's a mediation signature.

3.7 Moderation: price sensitivity weakens the effect

```
fit_mod <- fit_moderation_model(clean)
knitr::kable(
  broom::tidy(fit_mod, conf.int = TRUE) |>
  dplyr::filter(grepl("condition|price", term)),
  digits = 3
)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
conditionscarcity_frame	0.090	0.197	0.455	0.649	-0.298	0.477
conditionsustainability_frame	0.270	0.199	1.353	0.177	-0.123	0.662
conditionsocial_proof_frame	0.091	0.212	0.431	0.667	-0.326	0.509
conditioncombined_frame	0.475	0.210	2.257	0.025	0.061	0.889
price_sensitivity_c	-0.488	0.120	-4.076	0.000	-0.723	-0.252
conditionscarcity_frame:price_sensitivity_c	0.291	0.155	1.885	0.060	-0.013	0.596
conditionsustainability_frame:price_sensitivity_c	0.276	0.165	1.676	0.095	-0.048	0.600
conditionsocial_proof_frame:price_sensitivity_c	0.242	0.164	1.472	0.142	-0.082	0.565
conditioncombined_frame:price_sensitivity_c	0.354	0.170	2.076	0.039	0.018	0.689

```
plot_moderation(clean)
```

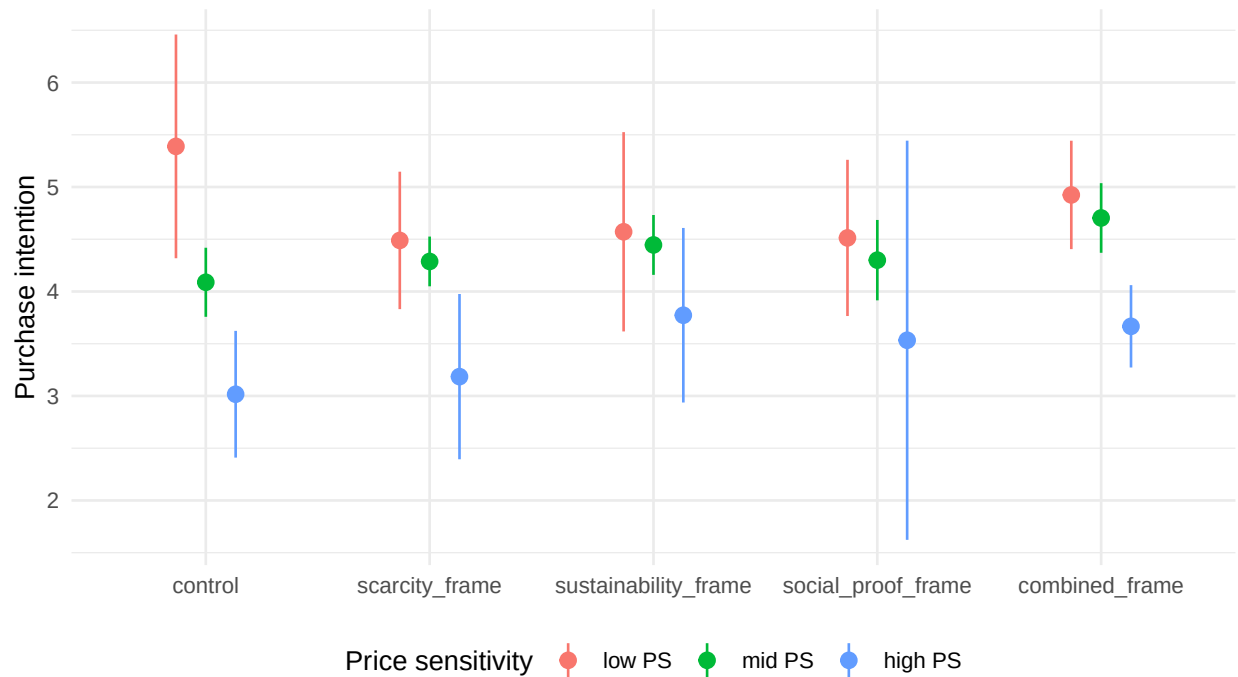


Figure 3: Condition $\times$ price sensitivity. The combined-frame boost shrinks for high-price-sensitive consumers.

4 Day 3 — Where does this go?

You're reading the Day 3 output right now. The same `.qmd` you're looking at becomes:

- **An HTML website** in `docs/`, served by GitHub Pages.
- **A PDF** ready to paste figures from into the LaTeX manuscript.
- **A Word document** for collaborators who don't use git.

4.1 The connectedness — what got reused, not copied

Every piece of this document came from somewhere else in the repo:

Element on this page	Where it lives	Used by who else
Variable names (<code>outcome_var</code> , ...)	R/01_setup.R	Every R script + Shiny app
<code>clean_consumer_data()</code>	R/functions/data_loading.R	<code>02_data_processing.R</code> , Shiny app
<code>fit_main_model()</code> ,	R/functions/analysis.R	<code>03_analysis.R</code> , manuscript
<code>fit_mediation_models()</code>		
<code>plot_pi_by_condition()</code> etc.	R/functions/plotting.R	<code>03_analysis.R</code> (figures saved to <code>figures/</code>)
Citations [<code>@baron1986</code>]	references/references.bib	This file + the LaTeX manuscript

Element on this page	Where it lives	Used by who else
Mock data	<code>data/mock/</code>	This file + Shiny app + LLM on Day 3

If you renamed `purchase_intention` to `intent_to_buy`, you would change it in `R/01_setup.R` and `R/functions/data_loading.R` — **two files** — and every figure, table, PDF, website page, and the Shiny app would follow.

4.2 What happens when you point this at real data

Set `DATA_MODE=real` in your `.Renvirom`. Re-render. Same code, same analysis, real numbers.

```
echo 'DATA_MODE=real' >> .Renvirom
quarto render reports/00_walkthrough.qmd
```

The mock workflow is the rehearsal. The real workflow is one env var away.

4.3 On Day 3 you also brought in the LLM

If you opened this repo with the LLM-mode workspace (`the-science-repository.llm.code-workspace`), your assistant can see this file and `data/mock/` but **not** `data/raw/` or `data/processed/`. That’s not a vibe; it’s enforced by:

- The workspace file’s `files.exclude` (editor-level hiding).
- `.claude/settings.json`’s `permissions.deny` (tool-level blocking).

So you can ask Claude to “improve the moderation plot” and it can do the job against synthetic data without ever touching the real one.

5 Where to take this next

- Add a new analysis: write a function in `R/functions/`, call it from `R/03_analysis.R` and from a new section here. **Don’t copy code.**
- Add a new outcome: extend `descriptives_by_condition()` and re-render. All tables update.
- Write the paper: open `manuscript/main.tex`. It uses the same `figures/` and `references/references.bib`.

6 References

- Baron, Reuben M., and David A. Kenny. 1986. “The Moderator-Mediator Variable Distinction in Social Psychological Research: Conceptual, Strategic, and Statistical Considerations.” *Journal of Personality and Social Psychology* 51 (6): 1173–82. <https://doi.org/10.1037/0022-3514.51.6.1173>.
- Hayes, Andrew F. 2017. *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*. 2nd ed. New York, NY: Guilford Press.